

ActInf Textbook Group ~ Cohort 7 ~ Session 1

(Chapter 1, Part 1) ~ 7/15/2024

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all right hello cohort 7 thank you everyone for all the introductions and and saying hello this is our first meeting so we've just all introduce ourselves and now we'll talk a little bit about chapter one slash anything active inference related kind of funneling into the textbook anything about the positioning of the textbook overall the two years since the publication and any topics that people are like excited about to explore in that third week which will give us like a little rest from the Rhythm otherwise um but let's go to chapter one and just like where does anyone want to begin what's a any whether it's a quote or section from chapter one or it's just something about their approach to active inference or it's a it's a general question uh that they have where where would anyone like to begin I'd like to ask a oh yes please please yeah so um um there's a statement on page five saying that active inference as a normative approach to understand brain and mind can you explain to me what normative approach means great great question I was wondering the same thing yeah yeah this is one of this is a this is a very important question I I I'm just going to where I I knew that the question had been asked but um Ali or anyone does anyone want to give a thought what what does normative mean when we're talking about active inference and a normative theoretical perspective yes ali uh well yeah there's an explanation for the use of this term normative in the notes and specifically on page um 267 I guess uh so the use of this term uh differs a little bit from its normal usage for example in philosophy literature or some in some other discourse uh so here the term normative basically refers to uh the way active inference modeling um somehow uh works works according to some evaluative standards so uh more precisely uh in active inference we have uh a specific parameter say surprisal or free energy which uh will get familiar uh more extensively in the uh coming chapters but uh there's this parameter around which all the other elements are evaluated uh and manipulated and modeled and so on so the word normative here actually refers to uh the framework of uh this Theory um and um it's not uh this is at least it's not exactly used in the same meaning as used in for example uh in ethics or in some I don't

know in other usage it's used in a in a rather technical way here so yeah thank you Ali yeah I added a link here to the social sciences course from last year and AVL and Ma work and Ben and ren too is discussing the social normativity social constraints and so that kind of like ought versus is and like what one ought or normed to do as Ali mentioned mentioned um but but it but it's we've explored that from some other directions too like to what extent does this outline how one ought to model Michael sure um yeah it is moving on to a different subject [Music] um uh I was kind of wondering uh so the agents or entities uh often have like a sort of highest preference with prior um I'm wondering if theoretically Andor practically I suppose these policies are always kind of externally artificially injected as like sort of vaccines or something like uh Fine Food say or if Act of inference is I guess I don't know if the word self-consistent um but if it these priors manifest automatically um by the I suppose perhaps uh notion of minimizing surprise or the uh the the bias priors are artificially injected every time yeah this is a this is a great question anyone have a thought on on it well normative even in this discussion is sometimes used in a very strong sense the core body temperature of a mammal has to be between a certain minimum and a certain maximum or that animal will stop moving that animal will die you have to have a certain amount of oxygen or you will stop moving and die uh there are some less violent impositions for instance anyone walking in public in a US city has to wear clothes and if you don't you will be taken away and therefore you will not be walking in a city anymore so then you can get into much more optional things like people have to be minimally nice to one another or people will stay away from you but uh there's a great range of uses uh all of which have the sort of ordinary language sense of ought to or have to must must not I think any of those things are norms of one kind or another and can be thought of normatively cool yeah I guess um you know they kind of help Kickstart and bootstrap the framing of a problem of Interest anyways otherwise how do you you have to Define boundaries at some point I suppose yeah yes the system of Interest choice and the boundary identification question in theory and practice this will come up a lot um Ali and then anyone else who raise their hand well yeah regarding Michael's question about uh how exactly prior SL preferences are implemented in the agent well I believe um it's both I mean it it can be implemented uh purely artificially in in an artificial way or it can arise from um as in activist would call through sense making uh so uh it can be pretty autonomous uh and uh it's not required uh to somehow implement those priors and preferences uh through some external implementation uh in self organ in self-organized agents and specifically in the organisms uh that participate in the sense making uh I I use this term uh in the inactive um inactive sense uh

those priors and preferences can arise from uh the very active sense making uh so yeah it can be I mean purely internal and autonomous as well thanks cool yeah there's so much to say on this like in the textbook we're learning about some of the Lego pieces and the the building blocks and some of the names of the colors and like that level of of fundamental and then we work in the second half of the textbook on some simple examples like listening to music and anticipation a rat in a tze some other kinds of of simple mapped cases like a pendulum swinging in physics and then the sophistication and the the complexity hierarchy is essentially open-ended because this is a cognitive modeling approach or systems modeling approach so when we're talking about complex territories then it's like a complex map so a lot of times there's this movements or tendency to to and and and an effective one at that to kind of like ground in a more basic example or Motif like something that we'll see in the textbook but then when we look at like a complex system like an organism or social setting then that's that's Str stretching the limits of our like mapping and modeling capacities and then we can kind of supplement that movement from the simpler Motif to the more complex system of Interest supplement that but okay well what what what is this new recent Ryan Smith paper on interoception like how are they modeling it here and so that helps enter in like more modern and empirical work otherwise we can go very simple look at the sort of Essentials that's uh and then we can also take it to um strive to model more complex systems any question or thought we can just or we can also look at some of the top prior questions so one feature that comes up in in um chapter one well there's two figures in chapter one this is the diotic phrasing of agent and Niche there's a deeper symmetry so it could be like agent agent Niche Niche it's just often represented as like a mobile agent in a world however the framework for interfaces boundaries blankets is more generic than that but this is kind of the two-part model with observation coming in action going out and then that is going to get nuanced into more of a four-part model where there's two things coming in and then there's two things coming out but we'll we'll get there so this is sort of just situating active inference within this ecological psychological tradition the cybernetic of agent and Niche and then we have the famous figure 1.2 which introduces and summarizes the low road and the high road this is a really important distinction that was introduced lightly in some earlier work it's referenced in a few of the questions uh about the desert landscape and here we have active inference as it's kind of like a metro map it's a schematic metro map so it's not like these are the only and the only ordering of terms along the way so don't commute with this map per se but it does describe the low road and the high road so-called so does anyone just like want to give any thought or overview

how did they see the the low road and the high road what distinguishes them how does this concept like come into play in the textbook um I don't know if this is a d point but I guess what interests me is somebody who doesn't know very much about bism at all but would like to is that it's interesting that they are too separate directions I guess naively I would have thought that the free energy principle was quite reliant on base theorem because of the definition of surprise so I guess it's interesting to me the idea that there are two roads and one doesn't kind of depend on the other but they meet at one place interesting thank you thean and then Frasier I I've looked into this I'm I'll make a comment but it might be a question um so my sort of take on things is that the low road is a perceptually based and that's distinctly different from the free energy principle side through predictive processing into active inference that is about um survival and resisting entropy and and that line of things so is that correct yeah that's that's great we well let's revisit this Fraser yeah just a thought on the low road and the high road to my mind I mean there was that great 15minute video of Carl from talking about the low road and the high road um as far as I can recall the the low road is perhaps you know we're starting with an idea about very very small uh micro states of some system and we're going to perhaps maybe this is a gas or something and we want to have a look at how those micro states are interacting and then through a kind of amalgamation of this you know the behavior of these micro States we can maybe get out of that an abstraction something like adaptive Behavior whereas the the higher road is a bit more like okay let's just take a bit more of a sort of an armchair approach and kind of you know conceptualize from a top down idea about what it would mean to be adaptive at all um kind of starting from an aggregate idea and then kind of going down and saying okay well we have this aggregate idea how are the micro States then going to have to behave so that that's kind of my thoughts on the the low road and high road distinction the only other thing I would like to mention is on that that initial figure you showed the agent observation loop I have in in various other uh parts of the literature cybernetics um cognitive science and so on heard this referred to as the the sensor motor loop I don't know if they actually used that terminology in the book I could be wrong it has been a little bit since I've actually read the book but that that seems to be a relatively Universal way to talk about this process of cyclical interaction between a system and its environment so I hope that's somewhat interesting thank you yeah so there's there's a lot of ways to talk about the in the high road um does anyone else want to just like give was any other thought on on low or high a textbook scale point to make is this sets up chapters two and three chapter two is going to take us from the low road to active inference chapter three is

going to take us from the high road to active inference and then chapter four is like the big like here you are here's the ACT inference generative model this is what you write in RX infer this is what you write on the napkin this is what you're sketching in the diagram like that's the active inference generative model which is both constructed along the low road path and compliant under the free energy principle umbrella so that's why the active inference generative model is sort of like the meeting point or the overlap because it's both a manifest artifact which is what allies it with these kinds of model architectures found all along the low road and it's a modeling output artifact which is compliant with this free energy principle and the basic mechanics so that's why sometimes people will talk about active inference as well it's a Cory it's like a downstream or it's a derivative of the free energy principle so it's like this is how you take free energy principle which we'll go into and apply that to persistent systems or adaptive or dynamic systems or taking the low road like cobbling and tinkering together like if you start with Bayes theorem now the last name does basically no work here Bayes theorem is just about conditional probabilities for example between observations of data and abductive inferences to latent States but we have here like if you wanted to build an architecture for perception and action even the special case like just a perceiver or just an actor if you wanted to do that in a fully synthetic setting like in silico like building agentic AI or building an API call or something like that or if you wanted to use this more of like an empirical cartography sense like making a model of a sensory motor system you're going to start implementing these different kinds of modeling choices and so built things are all along the low road which looks a lot in many cases like machine learning artificial intelligence these neural networks linear algebra stochastic dynamical processes and building up towards artifacts that contain this unified approach to perception action and one note on that coming only from the low road here not bringing in fvp per se is in Signal processing which is dealing with that inbound perception question action is often neglected and then on the contrary in control theory like balancing the plate on the stick and Mountain car all that in control theory the perceptual question is often neglected and so one piece of the low road approach here is taking a unifying approach to perception action not erasing their differences but having them as part of a unified imperative evidence maximization surprisal minimization bounded by the variational free energy so that's kind of one element is having perception and action in a unified model and then kind of a broader commitment to the unifying of all these diverse cognitive phenomena like attention memory learning anticipation all these different kinds of phenomena which otherwise it would be totally grab bag H how would these be linked and there could be like

infinite pairwise combinations and models that might not survive having a third piece of the puzzle added but by taking this kind of broader unificationist perspective the the whole iguana perspective like we're going to use an integrated toolkit to model the whole thing in its phenomena as it appears to the modeler that's what gives Unity on the low road and then that's without getting into the free energy principle and and what it brings from that first principles perspective then oh sorry or anyone else who wants to raise their hand or or give a thought um yeah I just add one thing to your your comment on the the grab bag of like attention and memory and everything like that because it's actually something we're working on within the marketing field right now um could you speak more to like how this this approach would help keep that from being that grab bag because that is basically the way it works in a practical sense at least in my industry yeah great question what does it really look like or mean for active inference to be a unifying approach to modeling different cognitive phenomena what would anyone say yeah Fraser hi oh I'll let uh someone else go I've talked already I heard someone sorry that was me did did you want to go Tom or uh well I I I guess um and this so I'm sure you'll have a smarter thing to say but um I guess my understanding of a lot of these sort of predictive processing models was that there's no difference really between a sort of a desire and a prediction and so the sort of this sort of classical psychology state where you have a want and you act on the world and you know what your Desir is like and you know when you've kind of reached that goal because you're like what you reserve the reality matches what your prediction was is sort of collapsed so that free energy sort of replaces like through surprisal replaces this old distinction between like what a desire is and what a prediction about the world is so my understanding is that those are united there and the attention is seen more as a sort of I guess a process by which that's achieved as in when you attend to something you're reting your prior about it but I could be wrong there um that could all be nonsense but yeah frer thank you Tom I mean yeah that that is in essence what I was going to to recapitulate I mean the so the initial question was how might active infants bring a sort of unifying approach to cognitive phenomena in general I think that was the question um I I'm a very very big fan of the work of uh Dr John V at the University of Toronto is's a cognitive scientist so his kind of whole Obsession in life is this idea of how it is that um we maintain adaptivity per se as an agent what makes us intelligent and he thinks it's this this process called relevance realization we're able to zero in on the relevant information essentially um so all cognitive processes under this idea would have to do with ability to zero in on the relevant information and what what might that be well in this case we have this framing of a sensor motor Loop

you know this econ Niche that we're constructing for ourselves to maintain our our integrity across time so I guess in in the in the broadest possible conceptualization you know active INF is a way of talking about how the adaptivity of that sensor motor Loop is is tuned uh in order to result in the perpetuation of the sensor motor Loop so and what you have to do what that boils down to is what Tom said basically is you have to have a you have to embody a model of what your world is in order to be able to uh match and and be in sync with it so that you can you can avoid things that are dous to your your boundaries and you can you can take in things that are good for you and so on so I'll stop talking he yeah there there's a lot to say and I and I I think we'll also be able to pick out some uh things that the that the authors the textbook say I'll give kind of a a a just a a primer response on this which is to go to figure 4.3 so we're looking ahead to what an active inference generative model can look like in this model whether looking at it in the discrete time or in the continuous time setting there aren't like advanced cogn there isn't self-reflexivity there's not like material reconstruction so there are phenomena in the real world in the real territory that are not in every single active inference map part of the cool thing about active inference is you can apply it to a rock not that it's the best tool but you could you could apply it to um a cloud as it kind of coalesces over a mountain or something like with a rain shadow so it doesn't need to apply to Any Given system or cognitive system and so because of that generality one way to think of it is active inference says less as a technique which enables people to say more and different about systems of Interest so how does this relate to how different psychological phenomena cognitive phenomena get unified in practice so let's just say that somebody says well I'm interested in the relation ship between sensory Precision prior beliefs and agency and estimates of efficacy in the world well without a first principal's approach that could be just a mega open-ended literature search you're going to hit on every single paper and journal on all those and then you know what another way would be to say okay well I'm going to associate sensory Precision with the a matrix here I'm going to associate prior beliefs and their possible differences with d and with efficacy I'll associated with b and I'll do this and this and that and so I'll kind of compose my Legos together and then I'll do model sweeps to explore the accuracy minus complexity of model fit like model preference and efficacy for this empirical data set which one of these synthetic models fits the data best so or maybe we just want to do the big one or or and test something specific about the biggest possible combinatoric model so even though this can initially feel like less of an answer what enables it to actually be more of an answer through time is that active inference isn't opinionated on what it would even mean

for attention to connect to agency but then in practice when you see a paper says by agency we mean this by attention we mean this and here's how we modeled their interactions so that's the unified compliant playground that allows modelers to actually specify and then have interoperability under the unified imperative so then models can be tested like with Factor one factor two Factor one and two and then there's actually a coherent way to say where one is more preferable and what data set they have what kind of performance on instead of getting into these like more ontological debates about you know well how does it happen in the brain well then that's great to zoom in and study how it really might be happening in the brain but that will never give a generic answer because now we're looking at a system of Interest Tom thank you this is a very basic question um so this idea is like you know you might have like one Matrix that means attention or it mathematically instantiates this idea does that mean you could create an organism with some capabilities like some kind of cognitive capabilities in the model that don't exist in the real world you know so you're not you don't say as a framework you need to have some mathematical bit that does attention some mathematical bit that does prediction but it that it's generic enough that you could instantiate really any kind of thing that you could receive as a cognitive capacity yeah that's a great way to put it um one thing that Maxwell ramstead said in the Discord that I remembered a few weeks ago was the focus on agents has led people to underestimate the scope of free energy principle basic mechanics so not just is AC of inference unopinionated on cognitive architectures of Any Given agent it's unopinionated on how one might distinguish or model even or even need there to be agents you could just make a fabric of API calls or you could make any or you could make a fantasy subway map that doesn't need to have any resemblance to any known cognitive system or any extant system at all again will that be useful for this or that will somebody pay you for this or that will that startup work I mean those are secondary questions but when we see that well we're talking about like base graphs and then we're going to associate incoming directionality with perception and outgoing with action but that doesn't need to have any kind of that's that's I'm not going to say it's the limits of what can be said there but that is the toolkit and then someone can make an architecture that has any kind of existence within an arbitrary Niche especially if it's in an immortal setting like the Mortal Computing that's where it gets into the the strange Loop and the embodied and the survival and and that's a very interesting Direction but especially in silico where the survival imperative is just the modeler's Whimsy you can make arbitrary things and making simple pedagogical things is often a a totally critical step like when we look at the the music modeling case in chapter 7 this this isn't claiming to a total

model of Aesthetics or of culture someone says yeah but also they they might have to breathe where's the oxygen in this model it's like it's just one facet or Motif that makes sense thank you yeah it's really fun to to explore like what are the limits and the constraints of modeling of formalization of experimentation and then within that how does active inference differ from near relative approaches so there's like kind of like a continent scale and then a more fine scale differentiation sometimes like are we comparing active inference with some unnamed or names qualitative approach which is a little bit like an apple and an orange comparison which isn't necessarily a bad thing just they're two different kinds of stances or are we comparing active inference to a model where perception and action are not considered with a unified imperative but then let's see what that model is and then that's more of a model to model discussion so be the difference between two competing hypotheses or two alternative models on a data set versus the question that's broader which is like should we model this data set at all cool well I hope this is a little flavor of chapter one and and some fun people who we'll be spending the coming months with again if you want to join the 13 UTC we'll be in chapter 6 which is super fun and a great starting point as well which is the recipe and that's the steps that we'll really go through to make active inference models in cohort 6 um or otherwise we'll catch up and uh Andrew Andor Ali or maybe others will be facilitating but um in the time between people can add questions add and augment just if you want to do something and contribute to it and you don't know what again just let blanket know okay cool thank you till next time thanks everyone bye thank you everyone thanks everyone